ARDMS VASC CTL: US Guided Procedures 7%

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Correct Answers in Red | Incorrect Answers in Red

1. At the start of an endovenous ablation procedure for the great saphenous vein, the catheter is inserted _____ and advanced to a point _____ the saphenofemoral junction. Points: 0 / 1

- the distal thigh, just above the saphenopopliteal junction
- with the IJV at the groin, at the most distal point of reflux
- the distal thigh, just above the saphenopopliteal junction
- the IJV at the groin, at the most distal point of reflux

You answered: B
Correct answer: C, D

Feedback
Proximal to the saphenofemoral junction

2. If the peak velocity in a transjugular intrahepatic portosystemic shunt is 150 cm/sec and the velocity in the hepatic vein and the vena cava is 35 cm/sec, what is indicated? Points: 0 / 1

- This mid-grade velocity is very low and indicates impending failure of the shunt.
- This velocity is normal. Normal flow within the shunt.
- There is mild flow acceleration in the graft at the proximal end.
- There is approximately a 50% stenosis at the proximal anastomosis.

You answered: A
Correct answer: A, B

Feedback
Flow velocity, reflective to the venacavopic shunt indicates impending failure.

3. Which of the following techniques **cannot** be used to monitor dose delivery during a transcutaneous radiofrequency ablation for a pseudoaneurysm is performed? Points: 0 / 1

- PRV sensor on the ground
- Doppler evaluation of the PV or SMA
- Pressure sensors on the graft lid
- Pulse oximeter on the ankle

You answered: D
Correct answer: C, D

Feedback
Pulse oximetry can be used for identifying the presence of the radial artery in the tissue. PW and CW Doppler can be used to monitor the flow and it is the best technique with the patient entirely based on the radial artery pulse.

4. Which of the following is a treatment for a pseudoaneurysm? Points: 0 / 1

- Compression sonotherapy
- Percutaneous injection
- Coagulum
- Stimpore

You answered: B

Feedback
A Thrombin injection is used to treat pseudoaneurysm with large necks. Ultrasound guidance is used to make the injection that is performed under sterile conditions. Thrombin is a hemostatic agent that promotes the formation of a blood clot. Thrombin is added to the hematoma or the pseudoaneurysm. Stimpore is a product of Medtronic.

5. A patient is referred to the vascular lab to evaluate the placement of a Greenfield filter. Where will you look for this? Points: 0 / 1

- abdominal aorta
- IVC at the left groin
- IJV
- subclavian or internal jugular vein

You answered: C

Feedback
The filter is placed in the IVC below the level of the renal veins to prevent any clots from the lower extremities from reaching the lungs.

7. Which of the following is an advantage of intraoperative vascular ultrasound compared to preoperative vascular ultrasound? Points: 1 / 1

- Technique does not require sterile ground or surgical field maintenance
- Can assess flow in all branches of the microvascular vessels
- Provides real-time flow evaluation
- Can be performed during the surgical procedure

You answered: C

Feedback
Angiography is an invasive procedure that requires a sterile ground and surgical field to administer contrast. Intraoperative US is performed by the surgeon in real-time and can be performed during the surgical procedure. Intraoperative US is a non-invasive technique and does not provide real-time flow information because the contrast flow is enhanced but with fluoroscopy or angiography. Intraoperative US is performed in the operating room and does not require a sterile ground or surgical field. Intraoperative US is not done in the OR. Bed-side ultrasound can be performed during the surgery.

8. Which of the following statements best describes the Doppler tracing found in the walls of a pseudoaneurysm? Points: 1 / 1

- high resistance to and fro flow
- low resistance to and fro flow
- low resistance to and fro flow
- continuous forward flow

You answered: A

Feedback
The neck of the sac with a narrow neck demonstrates a low resistance "swirling" of flow on the Doppler tracing. The necklace of the pseudo aneurysm is normally very narrow and will demonstrate a high resistance, turbulent waveform.

9. What is the preferred patient position during catheter retraction in an endovenous laser ablation treatment? Points: 1 / 1

- supine with legs abducted
- reverse Trendelenburg
- Trendelenburg
- standing

You answered: C

Feedback
When the catheter is inserted at the start of the procedure, the semi-erect or reverse Trendelenburg position is helpful to better visualize the GSV and the SFJ. The patient is placed in the Trendelenburg position during the catheter retraction. The Trendelenburg position is helpful to reduce the blood volume in the veins of the leg and facilitate the closure of the vein during the procedure. The patient is placed in the supine position with legs abducted during the endovenous laser ablation treatment. The patient is placed in the semi-erect position or standing for the ultrasound exam.

10. Which of the following is a contraindication for endovenous laser ablation of the great saphenous vein? Points: 0 / 1

- cellular incompetence in the great saphenous vein
- chronic obstruction of the ipsilateral vein
- tortuosity of the distal femoral vein
- Grade 2 reflux in the ipsilateral superficial femoral vein

You answered: B
Correct answer: C, E

Feedback
If a Grade 3 SEPHVENT is evaluated, the venous outflow system of a patient prior to venous ablation of the saphenous vein. If the venous outflow system is obstructed, the blood flow will be redirected to the deep venous system and will cause the venous hypertension.

11. Which of the following requires the use of sterile technique? Points: 0 / 1

- cast abatement for osteogenesis
- Doppler exam
- Sclerotherapy venous ablation
- Sonohysterog

You answered: D
Correct answer: C, E

Feedback
Sterile ground and surgical field maintenance is required for sonohysterography and sclerotherapy. Sterile technique is not required for cast abatement and Doppler exam.

12. What is an absolute contraindication for sonohysterog? Points: 0 / 1

- polycythemia vera
- vein location above the inguinal ligament
- vein location immediately over a lung prosthesis
- patient allergic to iodine

You answered: D
Correct answer: A, B

Feedback
Relative contraindications are situations when risks a particular treatment or procedure obviously outweigh. The risk of the procedure is greater than the benefit. The patient allergic to the iodine contrast agent is a relative contraindication. The patient with low platelet count is a relative contraindication.

13. Which of the following is true regarding the initial post-op evaluation of a transjugular intrahepatic portosystemic shunt placement? Points: 0 / 1

- A muscular perforator is the most frequently encountered complication.
- CW is most commonly performed for the initial evaluation of a TIPS procedure.
- Hepatic venous flow should be monophasic and pulsatile in the hepatic vein.
- A Doppler is needed in addition to venous flow to assessing blood volume.

You answered: C
Correct answer: A, B

Feedback
When performing and evaluation of a transjugular intrahepatic portosystemic shunt, it is important to establish a baseline for the patient's hepatic venous flow. The hepatic venous flow should be monophasic and pulsatile in a patient with hepatic congestion, cirrhosis and chronic pancreas.

14. A patient presents for ultrasound evaluation of a PICC line. The hub is located on the shoulder and lower extremity of the patient. Points: 0 / 1

- Central catheter demonstrated in the right internal jugular vein
- Central catheter demonstrated in the left internal jugular vein
- Ultrasonic Doppler evaluation of the catheter in the right internal jugular vein
- Ultrasonic Doppler evaluation of the catheter in the left internal jugular vein

You answered: A
Correct answer: C, E

Feedback
The image demonstrates the longitudinal view of the catheter in the subclavian vein on the posterior image. The transverse view of the GV demonstrates the catheter in the internal jugular vein. The catheter is located in the left internal jugular vein.

15. Which of the following indicates a balloon angioplasty of an 85% stenosis of the left common femoral artery? Points: 0 / 1

- an increase in the ABI of above 0.2
- increased damping in the reflux phase
- velocity ratio greater than 2.0
- decrease in the pulsatility index distal to the treated segment

You answered: B
Correct answer: C, D

Feedback
Acceptable of patency angiography, an ABI increase +0.15 and +0.06 diameter reduction in duplex ultrasound indicates a successful angioplasty. Increased damping in the reflux phase is a sign of a hemodynamically significant stenosis. Decreased pulsatility index distal to the treated segment is a sign of a hemodynamically significant stenosis.

16. Which of the following is **NOT** a part of the information required in an informed consent form for an interventional vascular procedure? Points: 0 / 1

- Expected time to complete patient consent
- Description of the procedure
- Common complications that have experienced complications after the same procedure at the facility
- Alternative procedures that the patient may experience

You answered: A

Feedback
Informed consent process must include a description of the procedure and the list possible complications that the patient may experience. The patient must be given a chance to ask questions and understand the procedure.

17. When Thrombin is injected in a pseudoaneurysm, the clotting should be begin: Points: 0 / 1

- within 24 hours
- after 1 minute of manual compression following the injection
- within 5-10 seconds post- injection
- within 1-5 minutes post- injection

You answered: D
Correct answer: C, E

Feedback
Thrombin administered in preliminarily injecting the minute agents into a pseudoaneurysm to clear the opening and not the neck of the pseudoaneurysm. Thrombin is an enzyme that converts fibrinogen to fibrin and promotes the formation of a blood clot. Thrombin is added to the hematoma or the pseudoaneurysm. Thrombin is injected into the pseudoaneurysm.

18. A patient is undergoing balloon angioplasty of the left common femoral artery. The physician just completed the first expansion of the balloon and asks you to check the flow with Doppler. Where is the most appropriate place to evaluate flow? Points: 0 / 1

- popliteal artery
- dorsalis pedis or posterior tibial artery
- proximal superficial femoral artery
- distal superficial femoral artery

You answered: D
Correct answer: A, B

Feedback
The physician is trying to review flow distal to the obstruction. It is most important to evaluate flow in the most distal artery in the lower extremity. Flow should be evaluated in the PT or DP.

19. Which of the following is true regarding conversion techniques used in the treatment of a pseudoaneurysm? Points: 0 / 1

- Compression should be performed for 20 seconds and with color Doppler evaluation in between compressions
- Distance equivalent to 1000joules/cm should be performed in three-minute intervals and if this does not clear the neck, the procedure should be repeated.
- Blood pressure should be controlled, performed positions on the legs and the neck to the pseudoaneurysm is closed.
- Thrombin injection is more effective than compression techniques in pseudoaneurysms with large diameter necks.

You answered: A
Correct answer: A, B

Feedback
Thrombin injection is more effective than compression techniques in pseudoaneurysms with large diameter necks. Blood pressure should be controlled during compression techniques to avoid changes in blood flow from exercise. Compression should be performed for 20 seconds and with color Doppler evaluation in between compressions.

20. In most cases, a peripherally inserted central catheter (PICC) line is placed in the basilic vein and advanced until the tip reaches: Points: 0 / 1

- cubital vein
- brachial junction
- proximal axillary vein
- brachiocephalic vein

You answered: B
Correct answer: C, D

Feedback
The brachial vein is the venous catheter is positioned in a lateral vein, tape secondary the distal 1/3 of the GV. Jugular venous catheter is placed in the internal jugular vein, tape secondary the distal 1/3 of the GV.

21. What is the most common site of stenosis in a transjugular intrahepatic portosystemic shunt? Points: 0 / 1

- hepatic vein anastomosis
- Billroth anastomosis
- hepatic vein anastomosis
- portal vein anastomosis

You answered: C

Feedback
Increased HT - hepatic lines to reach peak velocity / lower resistance / proximal deterioration

Decreased HT - hepatic lines reach peak velocity / higher resistance / distal deterioration

Increased pulsatility index indicates a proximal stenosis. There is pulsatility at the artery between and the arteries proximal to the stenosis. Pulsatility is lost distal to the stenosis.

Flow is the most distal to the stenosis. Flow is decreased. Flow is reduced. Pulsatility. In the artery the flow is the most distal to the stenosis. Flow is reduced. Pulsatility. Flow is the most distal to the stenosis.

22. When evaluating the left arm for potential placement of a central venous catheter, the sonographer should be sure to evaluate the area surrounding the subclavian vein. What should you do not? Points: 0 / 1

- evaluate the venous system in the right arm
- measure the largest collateral for potential pathway for the catheter
- evaluate the venously anomaly related for potential catheter pass
- left the physician the patient is not a candidate for catheter placement

You answered: B
Correct answer: A, B

Feedback
Normal veins are distended on the pre-procedure to use for central venous catheter placement. Extra is a venous or a venous of the adjacent vein. Tape secondary the distal 1/3 of the GV. Jugular venous catheter is placed in the internal jugular vein, tape secondary the distal 1/3 of the GV.

Normal veins are distended for catheter placement. These findings are usually going to change the arm point in placement of the catheter. Venous anomaly is a variation in the venous system. Venous anomaly is a variation in the venous system. Venous anomaly is a variation in the venous system

30. Which of the following describes a difference between endoscopic ablation of the gastroesophageal junction and of the small papillomas only?

Points: 0 / 1

- The ESD cannot be treated by endoscopic ablation technique
- The ablation is carried out using the argon plasma coagulation probe and the polypectomy snare for the treatment of the ESD
- The ablation is carried out using the argon plasma coagulation probe and the polypectomy snare for the treatment of the GJ
- In the case of the procedure, the patient is placed in the supine position for the ESD treatment and in the prone position for the GJ treatment

You answered: A

Correct answer: D, E

Feedback

The patient is placed in the supine position for the ESD treatment and in the prone position for the GJ treatment. The ablation is carried out using the argon plasma coagulation probe and the polypectomy snare for the treatment of the GJ. The ablation is carried out using the argon plasma coagulation probe and the polypectomy snare for the treatment of the ESD. The ablation is carried out using the argon plasma coagulation probe and the polypectomy snare for the treatment of the GJ. The ESD cannot be treated by endoscopic ablation technique. Endoscopic ablation of the gastroesophageal junction and of the small papillomas only is a procedure where the tissue is destroyed from the inner lumen before biopsy and resection